NAR Genomics and Bioinformatics, 2026, , 1–13

Supplementary Data for Layered representation diagnostics for ultra-short metagenomic reads

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Abstract

Supplementary figures and source tables supporting the representation-diagnostic analyses, including baseline, estimator, perturbation, parameter, redundancy and runtime audits.

Keywords supplementary data, metagenomics, representation diagnostics

Supplementary data overview

This supplementary file collects the visual audits and source-table preview that support the main representation-diagnostic manuscript. Figures S1–S3 evaluate baseline controls, empirical information proxies and sequencing-error-aware perturbations. Figures S4–S7 examine local mutation fraction, P-channel counterfactuals, MSP bin and weight sensitivity, and dimension-matched high-k compressed baselines. Figures S8–S10 summarize P/MSP redundancy, runtime, relation audits and the CAMI II marine anonymous-read stability probe. Figure S11 reports the mechanism-aligned 2×2 local-change audit and property-scaling sensitivity. Tables S1–S7 provide the CAMI II preview, seven-group contrasts, MI inference, factorial contrasts, scaling sensitivity, complete feature definitions and public WGS assembly accessions.

Supplementary Figures

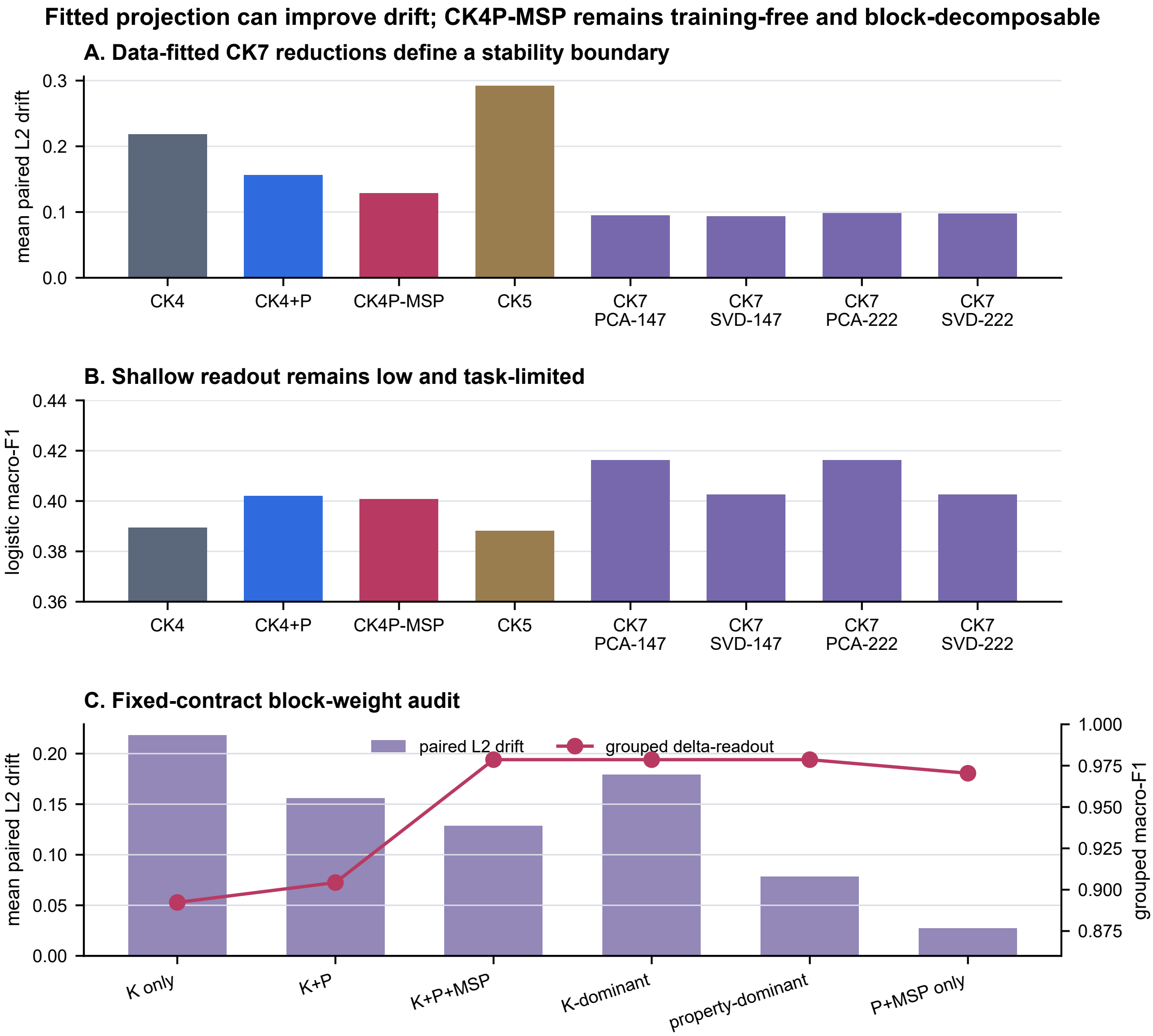

Figure S1 Fitted-reduction and block-weight boundary audit. (A) Mean standardized representation drift across 24 controlled WGS length-by-perturbation cells for CK4/CK5, CK4+P, CK4P-MSP and clean-partition-fitted CK7 PCA/SVD projections at 147 or 222 target dimensions; lower is preferable. The fitted CK7 projections achieve lower drift than CK4P-MSP and therefore define a data-dependent stability boundary. (B) Logistic macro-F1 across 132 WGS readout cells; projection rank in these smaller per-cell training partitions was sample-limited (median 49.5 components), so this panel is a shallow-readout boundary rather than a strict 147/222-dimensional comparison. (C) Mean drift across 24 WGS cells and grouped local-change macro-F1 across 12 cells under K-only, K+P, K+P+MSP and prespecified K/property-dominant weights. The identical grouped readout across the three non-zero K/P/MSP weightings shows that the default is not a fragile readout optimum.

Alt text: Three panels show that fitted CK7 PCA/SVD projections have lower drift than CK4P-MSP, all shallow readouts remain low, and the grouped local-change result is stable across non-zero K/P/MSP weightings.

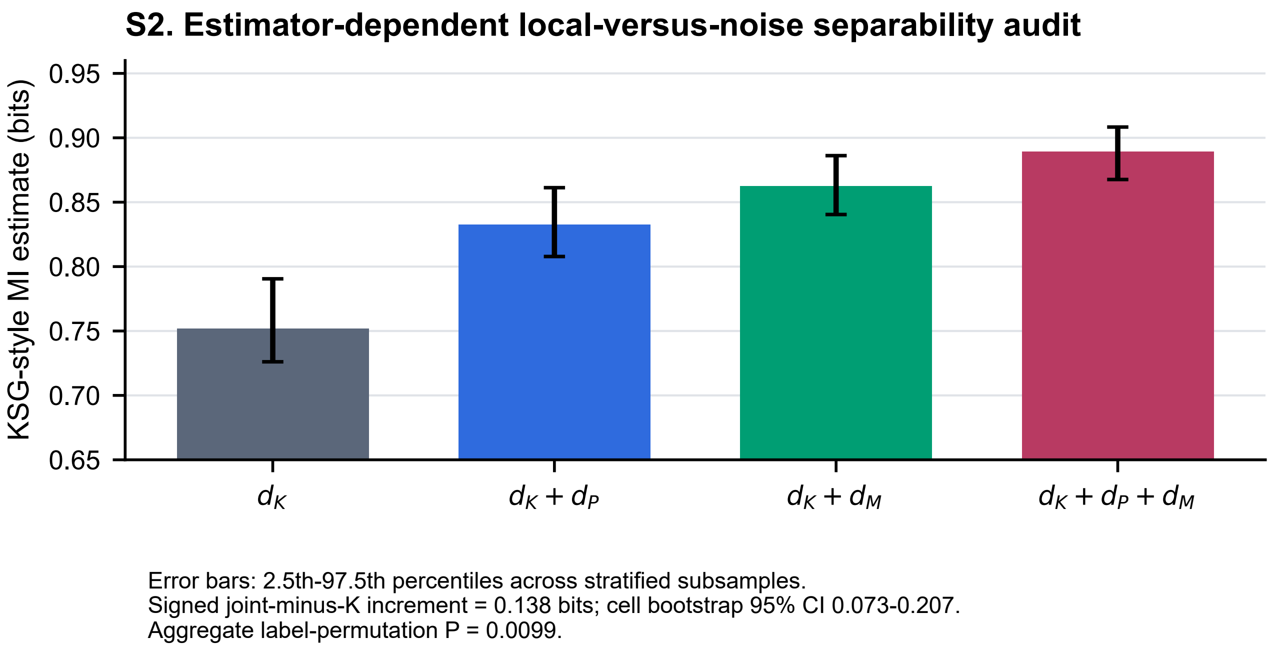

Figure S2 Estimator-dependent mutual-information (MI) audit for local change versus matched nuisance perturbation across 12 length-by-local-mode cells. Bars show KSG-style estimates for d_K, $d_K + d_P$, $d_K + d_M$ and $d_K + d_P + d_M$; error bars are the 2.5th–97.5th percentiles across 100 stratified subsamples per cell. The signed joint-minus-K increment was evaluated with a 10,000-resample paired-cell bootstrap and 100 shared label permutations per cell. Values are empirical separability estimates rather than universal information bounds; higher values indicate greater local-versus-nuisance separability under this estimator.

Alt text: Bar plots summarizing empirical mutual-information and conditional-mutual-information audit values across representation blocks.

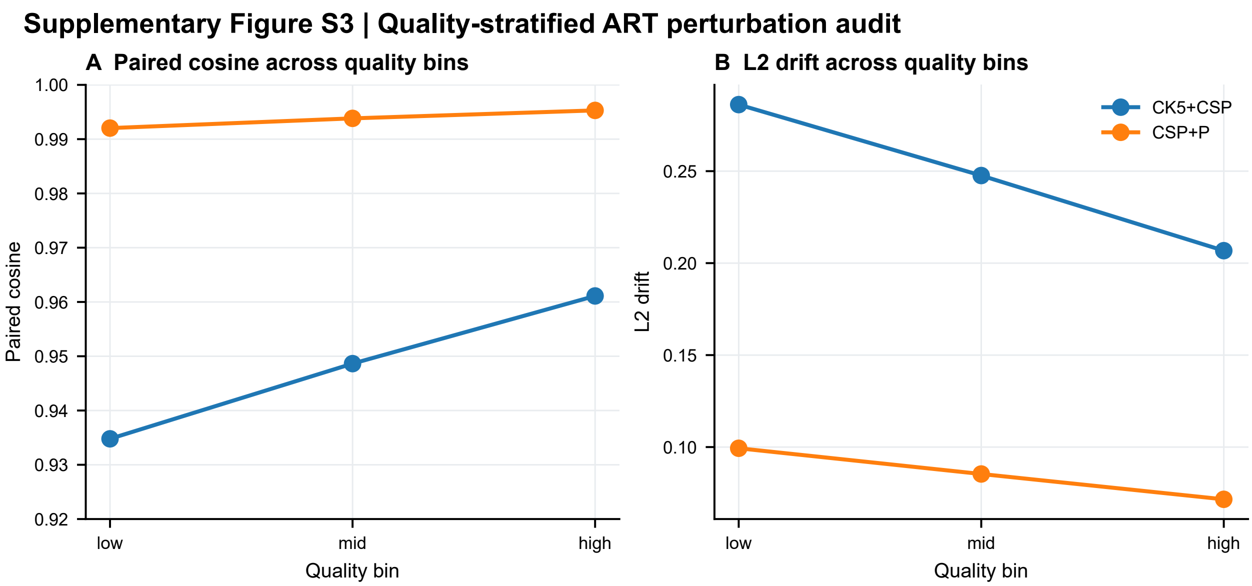

Figure S3 Quality-stratified ART contextual audit using 75,592 clean-versus-simulated rows from 37,796 source templates at 69, 75, 100, 125 and 150 bp. Panels summarize paired cosine and standardized L2 drift across low-, middle- and high-quality strata for the CSP+P and CK5+CSP comparators. Higher cosine and lower drift indicate greater stability. This historical mechanism probe does not contain a contract-v2 CK4P-MSP result and is used only to show that the simulator context was not evaluated as one pooled error source.

Alt text: Line plots showing ART perturbation behavior across quality levels for paired cosine and L2 drift.

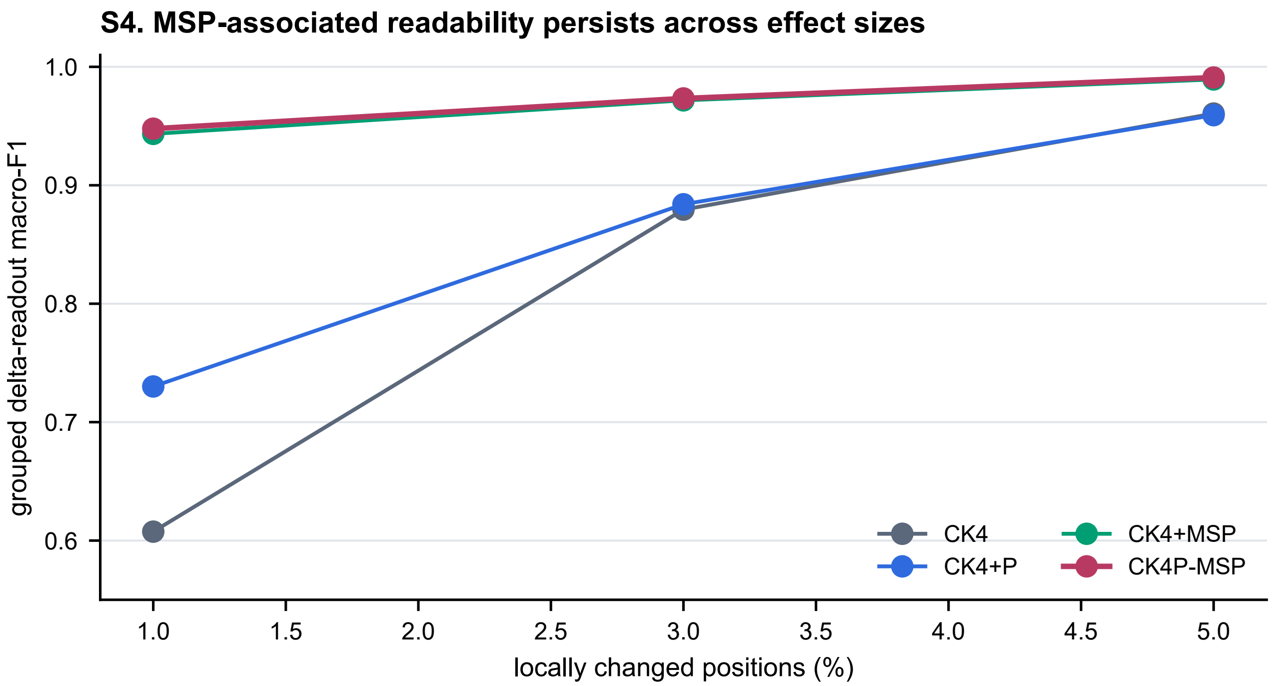

Figure S4 Contract-v2 local-change fraction sweep at 1%, 3% and 5% changed positions. Curves show logistic grouped delta-readout macro-F1 averaged across 12 length-by-local-mode cells, with 180 template triplets per cell and all derivatives of one template retained in one fold. Higher values indicate greater accessibility of local-versus-nuisance structure. CK4+MSP and CK4P-MSP retain the strongest readout across effect sizes; this synthetic mechanism probe is not a functional-variant classification task.

Alt text: Line plot showing local mutation-fraction sensitivity across compact representation families.

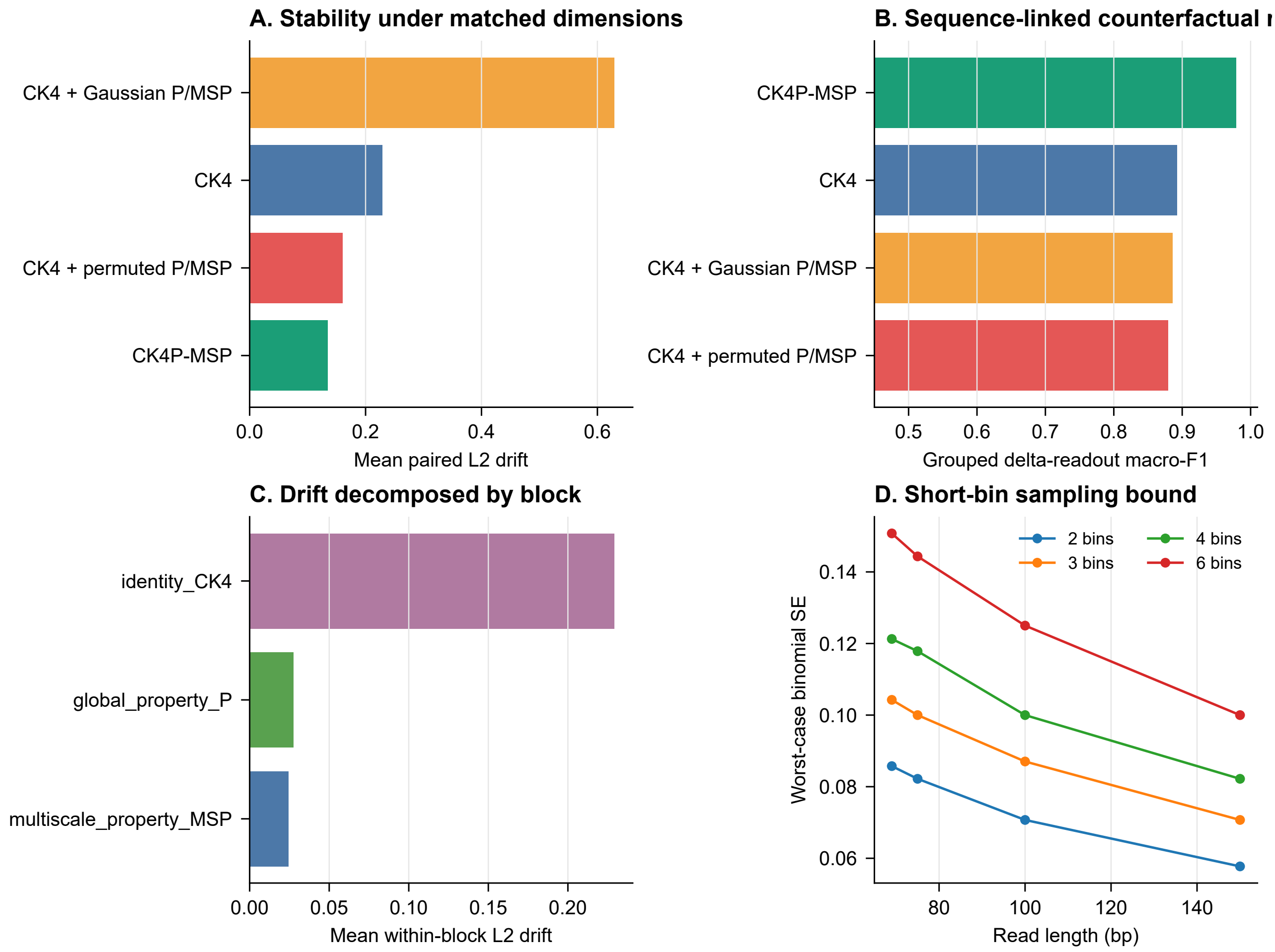

Figure S5 Sequence-linked P/MSP counterfactual and short-bin reliability audit. (A) Mean standardized drift across 20 WGS length-by-perturbation cells compares CK4P-MSP with CK4 and with row-permuted or marginally matched Gaussian P/MSP controls, all under the fixed block-normalization contract. (B) Logistic grouped delta-readout across 12 local-change cells (250 template triplets per cell) is higher for sequence-linked CK4P-MSP (0.979) than for CK4 (0.892), permuted P/MSP (0.879) or Gaussian P/MSP (0.886). (C) K, P and MSP blockwise drift separates their perturbation responses. (D) Worst-case binomial standard error is shown for 2-, 3-, 4- and 6-bin scales over 69–150 bp, based on the shortest bin at each scale. Lower drift and sampling error and higher readout are preferable on their respective panels.

Alt text: Multi-panel audit of P-channel counterfactual behavior, drift decomposition and short-bin sampling bounds.

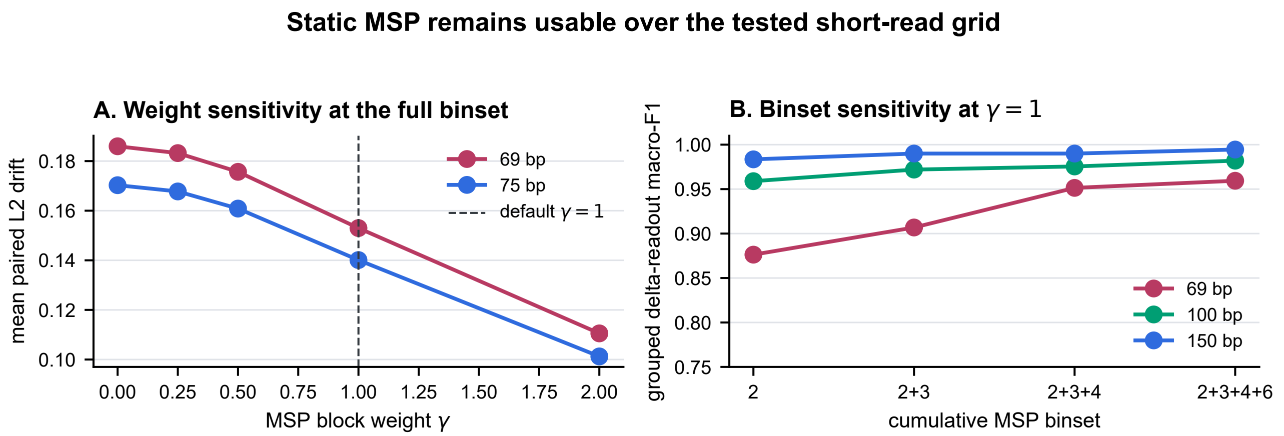

Figure S6 MSP binset and weight-sensitivity audit. (A) Mean drift over the prespecified WGS perturbations at 69 and 75 bp for the complete 2+3+4+6 binset and $\gamma \in \{0, 0.25, 0.5, 1, 2\}$; lower is preferable. (B) Logistic grouped delta-readout macro-F1 over 12 local-change cells at $\gamma = 1$ for cumulative binsets 2, 2+3, 2+3+4 and 2+3+4+6 at 69, 100 and 150 bp; higher is preferable. The audit tests sensitivity rather than selecting an optimal binset or weight.

Alt text: Line plots showing MSP binset and gamma-weight sensitivity for 69 and 75 bp reads and delta-readout.

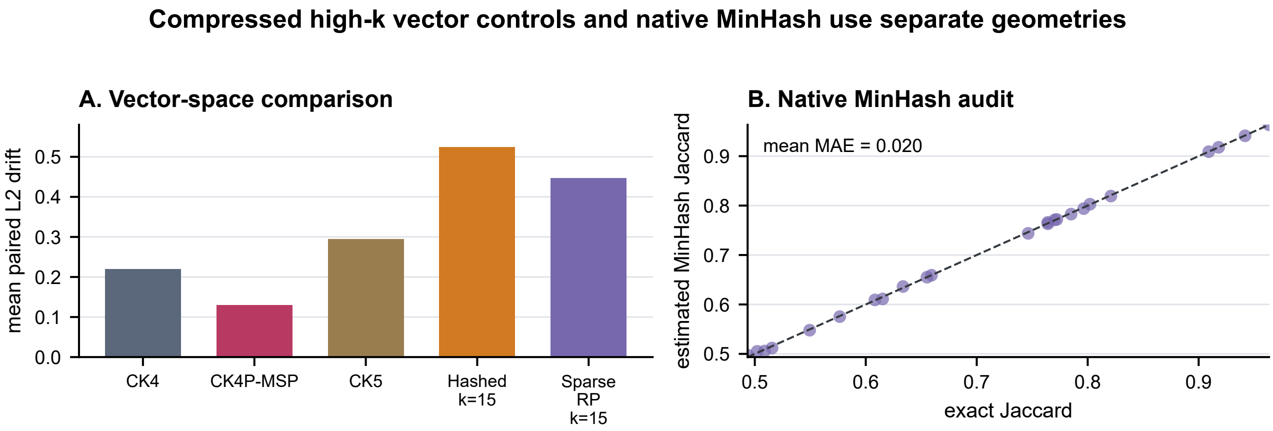

Figure S7 Dimension-matched high-k and native MinHash audit across 24 controlled WGS length-by-perturbation cells. (A) Hashed and sparse-random-projection canonical $k = 15$ vectors are compared with CK4, CK5 and CK4P-MSP at a 222-feature budget using mean standardized representation drift; lower is preferable. (B) Each point compares 222-hash MinHash signature agreement with exact $k = 15$ Jaccard similarity in one matched cell; the diagonal denotes equality and lower absolute error indicates better approximation. MinHash is not treated as an L2 feature vector.

Alt text: Multi-panel audit comparing dimension-matched high-k compressed vectors and native MinHash agreement with exact Jaccard similarity.

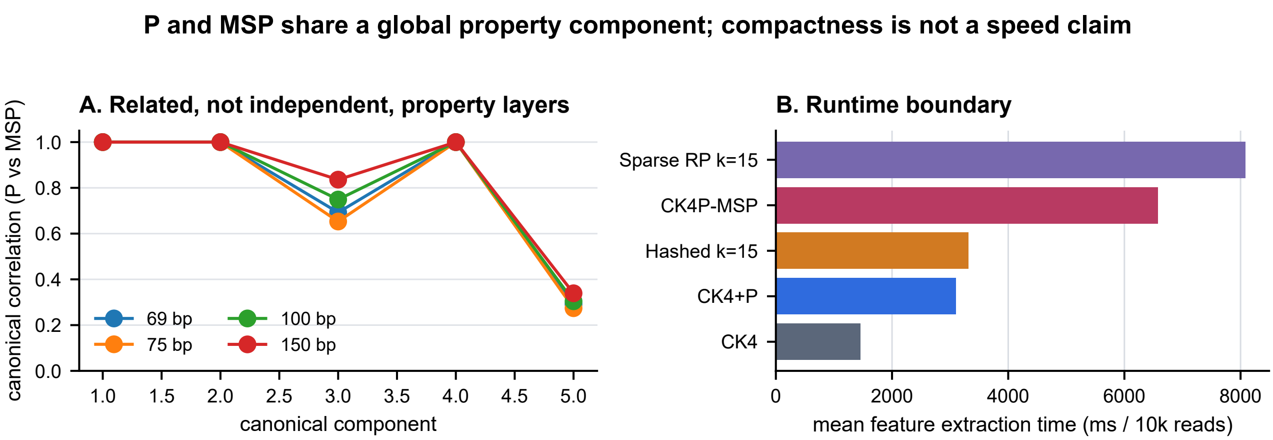

Figure S8 P/MSP relation and runtime audit. (A) Canonical correlations between P and MSP are shown across 600 clean reads at each of 69, 75, 100 and 150 bp; high early correlations indicate shared biochemical structure, whereas later components decline. (B) Mean feature-extraction time is reported from five repeated 10,000-read, 75-bp batches for each actual method path on the stated workstation. Lower runtime is faster. Timings are implementation- and hardware-specific and show that CK4P-MSP is compact in dimension but not the fastest encoder.

Alt text: Multi-panel audit showing P and MSP contributions, redundancy and runtime costs.

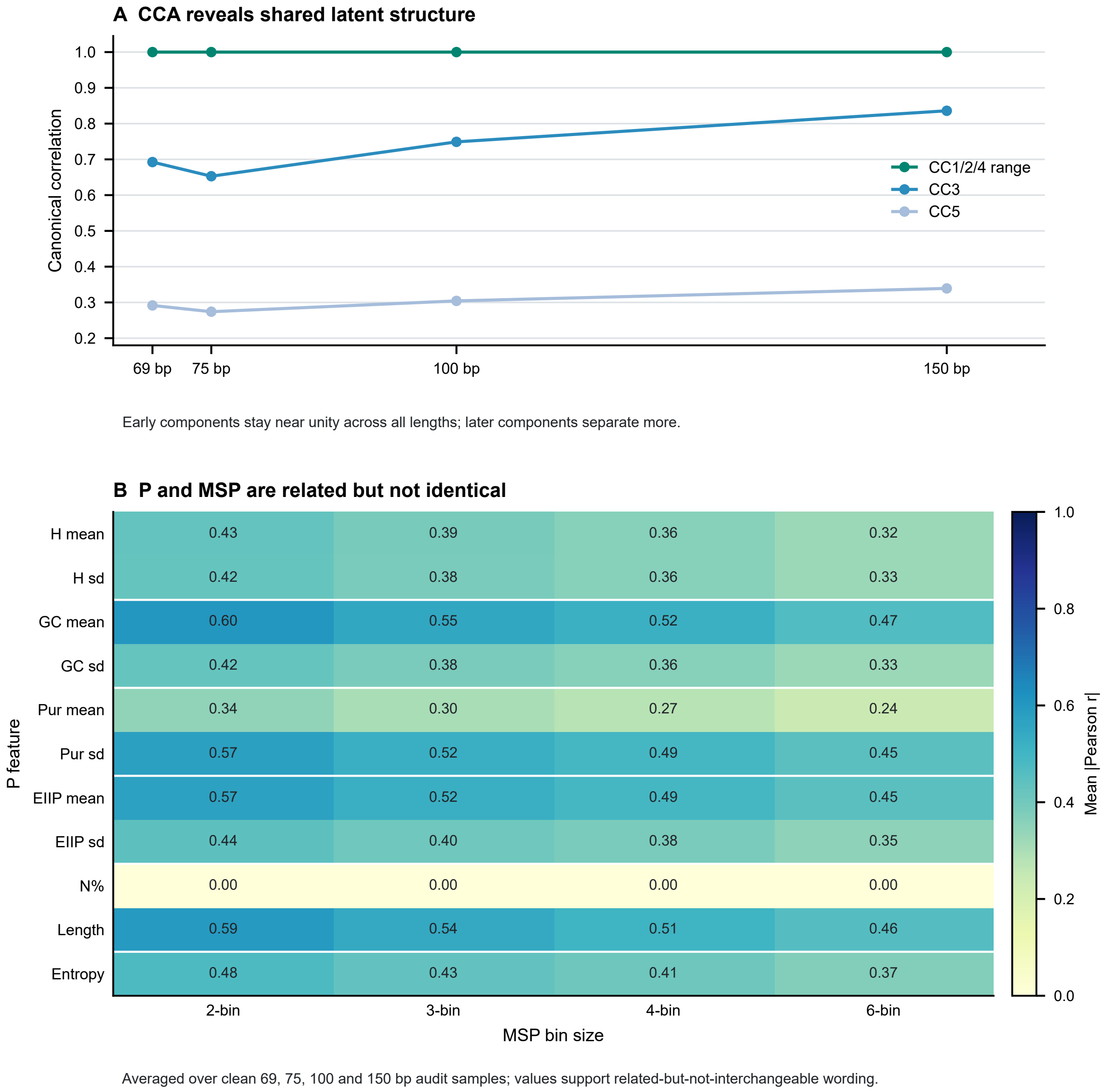

Figure S9 P/MSP relation audit using 600 clean reads at each of 69, 75, 100 and 150 bp. (A) Canonical-correlation analysis summarizes shared latent structure across the 11-dimensional P and 75-dimensional MSP blocks. (B) The heatmap reports mean absolute Pearson correlation between each P coordinate and each MSP bin-scale family, averaged across the four lengths. Values near one indicate strong association, not identity; the audit supports related-but-non-equivalent wording and is not a test of statistical independence.

Alt text: Line and heatmap panels showing CCA and correlation relationships between P and MSP feature layers.

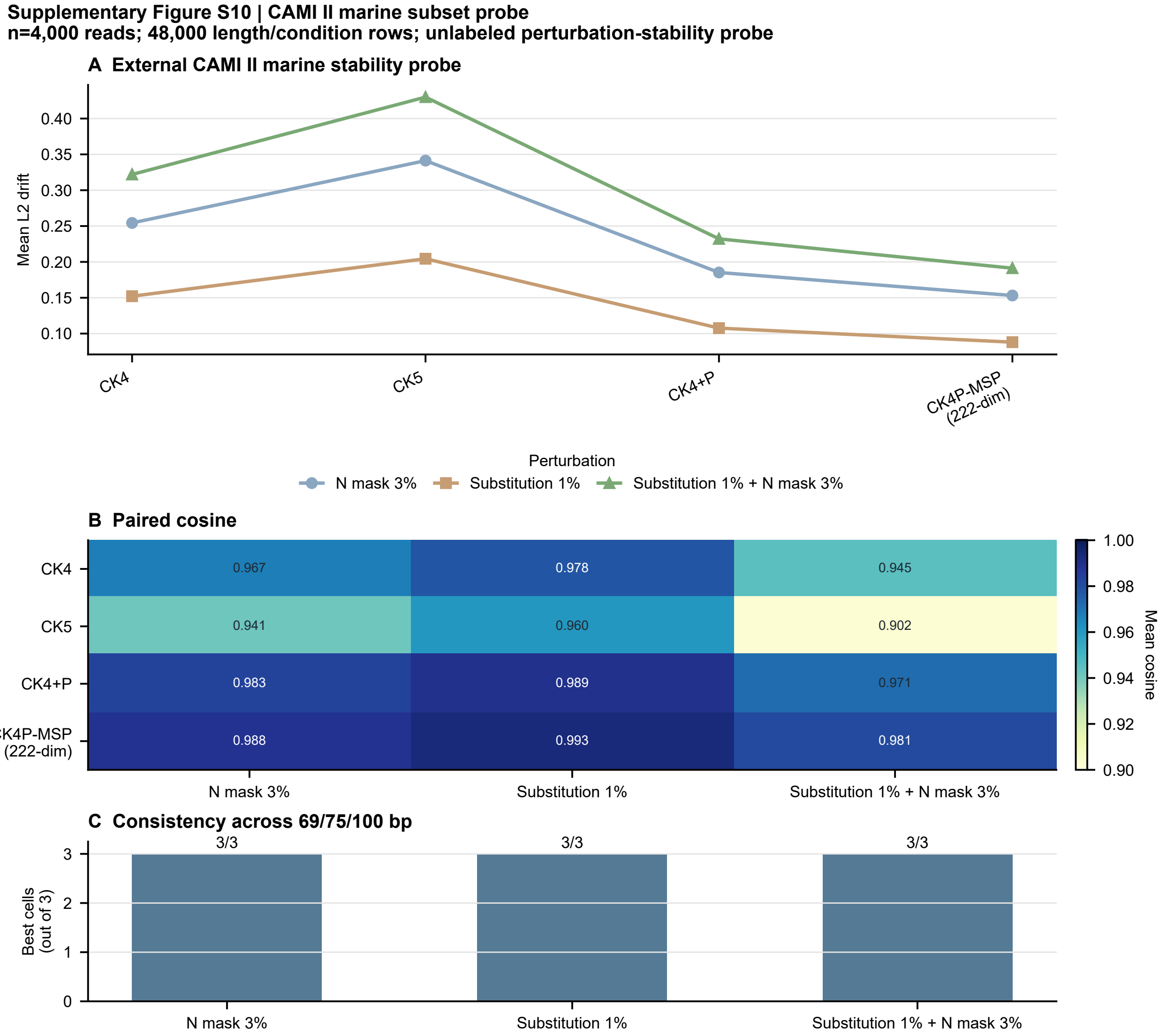

Figure S10 CAMI II marine anonymous-read stability probe. Paired cosine similarity, standardized L2 drift and nearest-clean retrieval are summarized for 4,000 anonymous source reads truncated to 69, 75 and 100 bp under N masking, substitution and combined perturbations. Read-level taxonomic labels were not reconstructed, so this figure supports external stability only.

Alt text: Line, heatmap and bar panels showing CAMI II marine anonymous-read stability across length and perturbation settings.

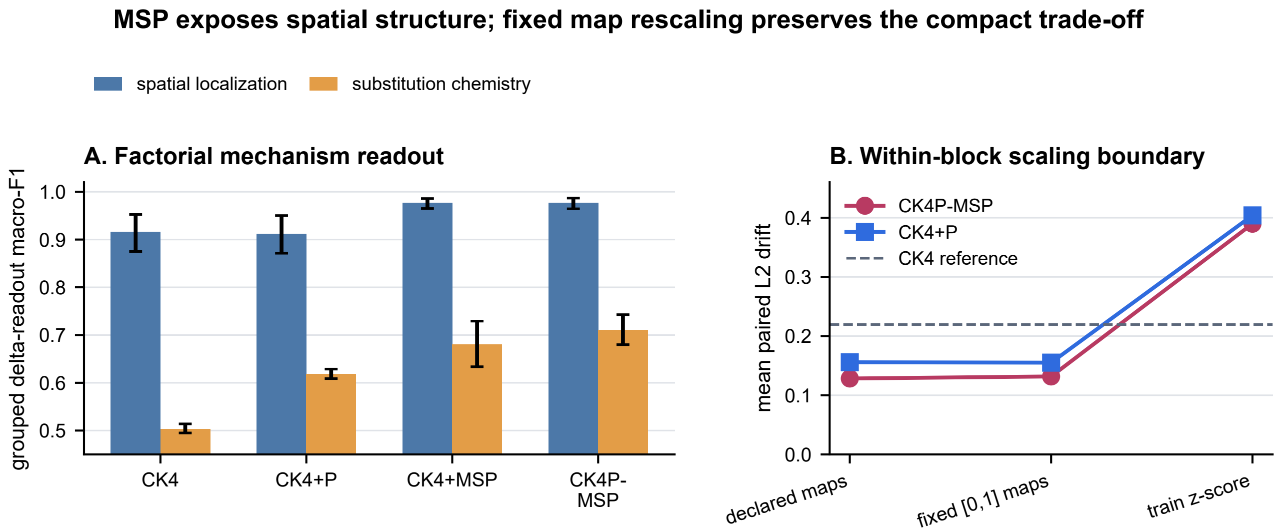

Figure S11 Mechanism-aligned local-change and property-scaling audits. (A) Grouped macro-F1 for spatial localization (contiguous versus dispersed substitutions) and substitution chemistry (property-directed versus random substitutions) across the four K-containing representations. Both tasks used 3,000 template-level observations per representation across 12 length-by-local-mode cells, with all derivatives of a template retained in one fold; error bars are 95% bootstrap intervals across cells. (B) Mean standardized representation drift across 24 WGS length-by-perturbation cells under the declared mappings, fixed per-property $[0, 1]$ mappings and train-fit coordinate z-scores. Higher macro-F1 and lower drift are preferable; z-scoring is a sensitivity analysis rather than an alternative default contract.

Alt text: Two panels showing that MSP improves spatial and chemistry readout, P adds a smaller chemistry increment, and fixed property remapping preserves drift while train-fit z-scoring does not.

Supplementary Tables

Table S1. Preview of the CAMI II marine subset source table. The full CSV is retained in the data package.

length	condition	method	features	paired cosine	L2 drift	top-1
69	N_3pct	CK4	136	0.9643	0.2651	1.0000
69	N_3pct	CK4+P	147	0.9944	0.1059	1.0000
69	N_3pct	CK4P-MSP	222	0.9948	0.1016	1.0000
69	N_3pct	CK4P-MSP+SD	297	0.9943	0.1061	1.0000
69	N_3pct	CK5	512	0.9382	0.3491	1.0000
69	N_3pct	CK5+CK4P-MSP	734	0.9665	0.2572	1.0000
69	N_3pct	CSP	147	0.9941	0.1080	1.0000
75	N_3pct	CK4	136	0.9683	0.2497	1.0000
75	N_3pct	CK4+P	147	0.9950	0.0992	1.0000
75	N_3pct	CK4P-MSP	222	0.9954	0.0951	1.0000
75	N_3pct	CK4P-MSP+SD	297	0.9950	0.0994	1.0000
75	N_3pct	CK5	512	0.9441	0.3322	1.0000
75	N_3pct	CK5+CK4P-MSP	734	0.9698	0.2444	1.0000
75	N_3pct	CSP	147	0.9948	0.1011	1.0000
100	N_3pct	CK4	136	0.9687	0.2483	1.0000

Table S2. Prespecified conditional-contribution contrasts from the seven-group K/P/MSP ablation. The full representation is CK4P-MSP. Improvement is signed so that positive values favour the full representation; for L2 drift, the sign is reversed because lower values are preferable. Confidence intervals are percentile intervals from 10,000 paired-cell bootstrap resamples. Two-sided Wilcoxon signed-rank tests were adjusted across all 12 rows by the Benjamini–Hochberg procedure. Global cells are length-by-perturbation pairs; local cells are length-by-local-mode grouped cross-validation summaries.

block contrast	evidence layer	metric	n	full	comparator	improvement [95% CI]; BH q
K | P+MSP	global	paired cosine	24	0.989	0.999	-0.010 $[-0.013, -0.008]$; 2.4×10^{-7}
K | P+MSP	global	L2 drift	24	0.129	0.027	-0.101 $[-0.118, -0.086]$; 2.4×10^{-7}
K | P+MSP	global	retrieval	24	1.000	0.942	0.058 $[0.036, 0.081]$; 7.3×10^{-4}
K | P+MSP	local	grouped macro-F1	12	0.979	0.970	0.008 $[-0.0002, 0.019]$; 0.507
P | K+MSP	global	paired cosine	24	0.989	0.984	0.005 $[0.004, 0.006]$; 2.4×10^{-7}
P | K+MSP	global	L2 drift	24	0.129	0.156	0.027 $[0.023, 0.031]$; 2.4×10^{-7}
P | K+MSP	global	retrieval	24	1.000	1.000	0.000 $[0.000, 0.000]$; 1.000
P | K+MSP	local	grouped macro-F1	12	0.979	0.978	0.001 $[-0.0002, 0.002]$; 0.525
MSP | K+P	global	paired cosine	24	0.989	0.984	0.005 $[0.004, 0.006]$; 2.4×10^{-7}
MSP | K+P	global	L2 drift	24	0.129	0.156	0.027 $[0.023, 0.032]$; 2.4×10^{-7}
MSP | K+P	global	retrieval	24	1.000	1.000	0.000 $[0.000, 0.000]$; 1.000
MSP | K+P	local	grouped macro-F1	12	0.979	0.904	0.074 $[0.035, 0.117]$; 7.3×10^{-4}

Table S3. Estimator-dependent empirical MI separability increment across 12 prespecified length-by-local-mode cells. The contrast is signed and was not clipped at zero. The confidence interval uses 10,000 paired-cell bootstrap resamples; the aggregate permutation P value uses 100 shared label permutations per cell.

contrast	n cells	mean (bits)	95% CI	Wilcoxon P	permutation P	k
$\widehat{I}(d_K, d_P, d_M; Y) - \widehat{I}(d_K; Y)$	12	0.138	[0.073, 0.207]	0.0004883	0.009901	5

Table S4. Prespecified conditional block contrasts in the 2×2 spatial-localization-by-substitution-chemistry audit. Macro-F1 was estimated by template-grouped fivefold cross-validation. Confidence intervals are 10,000-resample paired-cell bootstrap intervals, and q values are Benjamini–Hochberg adjusted across the six contrasts.

conditional block	readout target	full	comparator	$\triangle$ macro-F1 [95% CI]	BH q
K | P+MSP	spatial localization	0.977	0.964	0.013 [0.001, 0.025]	0.1553
K | P+MSP	substitution chemistry	0.710	0.758	-0.048 [-0.058, -0.037]	0.0009766
P | CK4+MSP	spatial localization	0.977	0.977	0.000 [-0.001, 0.002]	0.5195
P | CK4+MSP	substitution chemistry	0.710	0.680	0.030 [0.013, 0.048]	0.03149
MSP | CK4+P	spatial localization	0.977	0.912	0.065 [0.034, 0.098]	0.0009766
MSP | CK4+P	substitution chemistry	0.710	0.618	0.092 [0.059, 0.130]	0.0009766

Table S5. Property-scaling sensitivity across 24 WGS length-by-perturbation cells. The declared mapping is the manuscript contract; fixed $[0, 1]$ maps and train-fit coordinate z-scores test sensitivity to coordinate scaling.

representation	property scaling	paired cosine	L2 drift	retrieval
CK4P-MSP	declared maps	0.989	0.128	1.000
CK4P-MSP	fixed [0,1] maps	0.989	0.132	1.000
CK4P-MSP	train-fit coordinate z-score	0.887	0.390	0.987
CK4+P	declared maps	0.984	0.156	1.000
CK4+P	fixed [0,1] maps	0.984	0.155	1.000
CK4+P	train-fit coordinate z-score	0.869	0.404	0.840

Table S6. Complete P and MSP coordinate definitions. Population SD uses denominator L over the read-level mapped values. MSP uses bin means only in the primary representation.

block	coordinate family	definition	dimensions
P	hydrogen-bond class	Mean and population SD of A/T= 2, C/G= 3, N= 0	2
P	GC indicator	Mean and population SD of C/G= 1, A/T/N= 0	2
P	purine indicator	Mean and population SD of A/G= 1, C/T/N= 0	2
P	EIIP mapping	Mean and population SD of A= 0.1260, C= 0.1340, G= 0.0806, T= 0.1335, N= 0	2
P	ambiguity	Fraction of N bases among A/C/G/T/N characters	1
P	scaled length	Read length divided by 200	1
P	scaled entropy	Shannon entropy over A/C/G/T/N divided by $\log_2 5$	1
MSP	five per-base channels	Hydrogen, GC, purine, EIIP and N-indicator values; mean pooled within each relative-position bin	$5 \times 15 = 75$
MSP	relative-position bins	For each $b \in \{2, 3, 4, 6\}$, edges are $\lfloor jL/b \rfloor$; empty bins use a zero row; the main method excludes within-bin SD	15 bins

Table S7. Public reference assemblies used to sample the controlled WGS read templates. Each species is represented by the listed NCBI assembly accession.

genus	species	NCBI assembly accession
Acinetobacter	*Acinetobacter baumannii*	GCF_900011295.1
Acinetobacter	*Acinetobacter calcoaceticus*	GCF_000368965.1
Acinetobacter	*Acinetobacter nosocomialis*	GCF_041021905.1
Acinetobacter	*Acinetobacter pittii*	GCF_000369045.1
Burkholderia	*Burkholderia cenocepacia*	GCF_000009485.1
Burkholderia	*Burkholderia cepacia*	GCF_001411495.1
Burkholderia	*Burkholderia multivorans*	GCF_000010545.1
Candida	*Candida albicans*	GCF_000182965.3
Candida	*Candida dubliniensis*	GCF_000026945.1
Candida	*Candida parapsilosis*	GCF_000182765.1
Candida	*Candida tropicalis*	GCF_000006335.3
Enterobacter	*Enterobacter asburiae*	GCF_900077725.1
Enterobacter	*Enterobacter bugandensis*	GCF_900324475.1
Enterobacter	*Enterobacter cloacae*	GCF_905331265.2
Enterobacter	*Enterobacter hormaechei*	GCF_000213995.1
Enterobacter	*Enterobacter kobei*	GCF_023023125.1
Escherichia	*Escherichia coli*	GCF_000005845.2
Escherichia	*Escherichia fergusonii*	GCF_000026225.1
Klebsiella	*Klebsiella pneumoniae*	GCF_000240185.1
Klebsiella	*Klebsiella quasipneumoniae*	GCF_003181175.1
Klebsiella	*Klebsiella variicola*	GCF_000025465.1